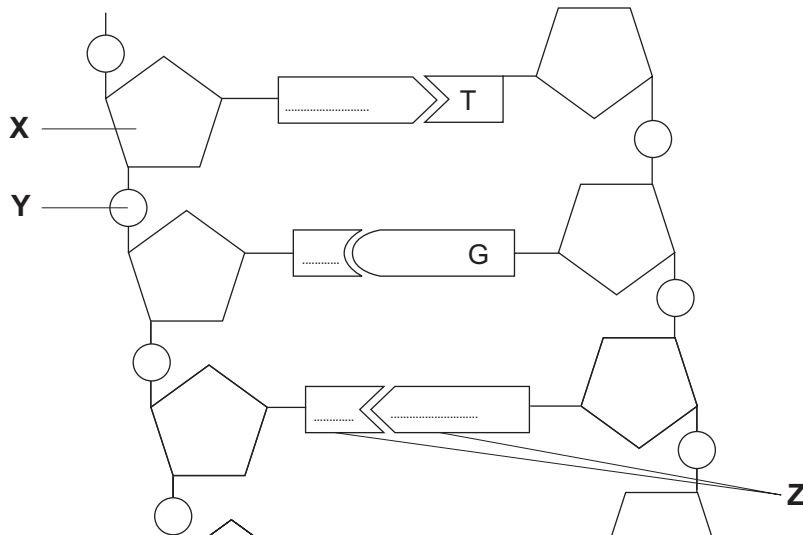


3. (a) The diagram below shows a small section of DNA.



- (i) Which one of the following pairs of substances correctly describes labels **X** and **Y** on the diagram?

Underline your answer

[1]

**salt and sugar**

**sugar and phosphate**

**acid and phosphate**

**salt and acid**

- (ii) The structures labelled **Z** are bases.

Complete the letter names for the **four** missing bases on the diagram.

[2]

- (iii) State why the order of the bases in DNA is important in the production of proteins.

[2]

.....

.....



(b) Apart from identical twins, no two persons have identical DNA. Samples of DNA can be analysed to produce DNA profiles which can be used to identify individuals in criminal investigations.

The diagram below shows five DNA profiles.

| DNA from<br>crime<br>scene | DNA from suspects |              |              |              |
|----------------------------|-------------------|--------------|--------------|--------------|
|                            | suspect<br>1      | suspect<br>2 | suspect<br>3 | suspect<br>4 |
| ████████                   | ████████          | ████████     | ████████     | ████████     |
| ████████                   | ████████          |              | ████████     | ████████     |
| ████████                   |                   | ████████     | ████████     |              |
| ████████                   | ████████          | ████████     | ████████     |              |
|                            | ████████          | ████████     |              | ████████     |
| ████████                   | ████████          | ████████     | ████████     | ████████     |
| ████████                   | ████████          |              | ████████     | ████████     |
| ████████                   | ████████          | ████████     | ████████     | ████████     |
| ████████                   |                   |              | ████████     |              |
| ████████                   | ████████          | ████████     | ████████     | ████████     |

(i) From the diagram, identify the suspect whose DNA was found at the crime scene. Give a reason for your answer. [1]

Suspect .....

Reason .....

(ii) Apart from criminal investigations, state **one other** use of DNA profiling. [1]

.....

(iii) After DNA profiles have been used in investigations they are often retained for future reference. Suggest **one** reason why some people may object to their DNA profiles being retained by the police. [1]

.....

